

Literature Review

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Overview of literature review

- Literature review of 13 papers
- Main topics: crack propagation in snow, snow modeling, modeling in more general sense (introduction to different modeling methods used)
- Modeling methods: Material Point Method (MPM) and Discrete Element Method (DEM)

Modeling Methods

- Two main methods used: MPM and DEM
- DEM described by Cundall and Strack, 1979, material of disks & spheres, contact forces and resulting displacements calculated
- MPM where material is discretized in points with Eulerian background mesh to calculate stresses at points and time-dependent equations, for example in Gaume et al., 2018

Bibliography



Cundall, P. A., & Strack, O. D. L. (1979). A discrete numerical model for granular assemblies. *Géotechnique*, 29(1), 47–65.

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Gaume, J., Gast, T., Teran, J., van Herwijnen, A., & Jiang, C. (2018). Dynamic anticrack propagation in snow. *Nature Communications*, 9(1), 3047.

<https://doi.org/10.1038/s41467-018-05181-w>



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